Fiscal Year:	FY 2018	**Task Last Updated:**	FY 03/23/2018
PI Name:	Narayanan, Ranga Ph.D.		
Project Title:	A Novel Way To Measure Interfacial Tension Using The Electrostatic Levitation Furnace (ELF)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
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A novel method using an Electrostatic Levitator (EL) is proposed to determine the interfacial tension (IFT) of liquid/gas systems. This key thermophysical property is of interest for materials processing and in welding operations in space and on ground. The novel method for measurement of IFT proposed involves the use of electrostatically forced Faraday instability. In this method, an oscillatory electrostatic forcing is applied to a liquid system, resulting in sudden pattern formation at the interface, i.e., an instability is caused. The "instability" is a result of resonance between the imposed forcing frequency and the natural frequency of the system. The critical amplitude and mode shapes are predictable from linear stability theory and agree superbly with experiments. The measurement of IFT directly aligns with the goals of the Thermophysical Property Measurement emphasis of the current NRA (NASA Research Announcement).
Faraday experiments by this team using mechanical forcing have shown excellent agreement between theory and

Task Description:	experimental results. However, in the case of mechanical forcing, the force of gravity overwhelms the effects of IFT on the system stability and therefore it is almost impossible to determine the IFT with accuracy. Electrostatic forcing counters gravity, allowing the effects of IFT to play a significant role in the ultimate determination of the system stability. Accordingly, experimentally mapped stability curves can be fit accurately with theoretical curves, allowing for the determination of the IFT with great accuracy. The proposing team has successfully used electrostatic forcing in ground equipment to show that Faraday waves can be obtained as per theoretical predictions. The current state of the art uses an open container geometry to hold the liquid. However, a levitated sphere may also be used, and our proposal necessarily includes this method. We therefore propose a project with two NEW aspects. In the first stage of the research, we propose to use ground-based facilities at JAXA (Japan Aerospace Exploration Agency) and NASA Marshall Space Flight Center (MSFC) to accurately obtain the IFT for liquids using electrostatic Faraday forcing in open containers as well as in levitated drops. In the second stage of the research, we will perform flight experiments using the Electrostatic Levitation Furnace (ELF) in the International Space Station (ISS) to obtain the threshold for instability and determine the IFT of these systems. In summary, the research allows an alternate new method, viz., Faraday instability, for the determination of interfacial tension. Using electrostatic forcing instability will permit an independent confirmation of measurements obtained by the single-pulse oscillating-drop technique where decay rates are measured and related to interfacial tension and viscosity. Ground-based research using the electrostatic levitator available at the NASA MSFC and JAXA, and space experiments on ELF in the ISS will provide accurate data. In addition to being an alternative method for IFT measurement, the proposed technique also has versatility as a potential means in measuring interfacial tension for high viscosity materials and two-liquid interfacial tension on ground as well as in space. The proposing team has recently collaborated successfully with JAXA and has conducted experiments to study the feasibility of electrostatic Faraday instability. These results form the basis for the current proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Accurate interfacial tension values are crucial in processes of importance to both industry and international space missions, including: a) Additive manufacturing of metals, where IFT knowledge is pivotal to avoid shrinkage and wrinkling. Human Exploration: TA7.2.3.5: Welding, TA12.1.2: Computationally-Designed Materials, TA12.4.2.5: Additive Manufacturing. b) Semiconductor crystal growth, where the IFT between a liquid melt and liquid encapsulant plays a large role in determining conditions where convection is present. c) Space enabling operations. One example is joining operations such as soldering, where the IFT plays an important role in the structure and stability of the weld. Accurate values for the IFT will enhance the safety and effectiveness of this type of operation in space where the microstructures formed are different than those on Earth where gravity is present. d) Enhanced oil recovery, where the IFT between injection fluid and crude oil plays an important role in the recovery rate of the oil and the prevention of viscous fingering. e) The accurate measurement of the IFT between liquids through the use of a different operational mode with minimal experimental modification can validate measurements made through the current techniques of levitation and pulse disturbance of spherical fluid masses.
Task Progress:	Two main tasks were accomplished in the first period. The first task is related to the measurement of surface tension using resonant or Faraday forcing in levitated, high temperature, liquid-metal drops. Resonant frequencies obtained experimentally have been correlated to surface tension measurements using the Electrostatic Levitator (ESL) facility at MSFC. Excellent agreement of surface tension with published values was obtained. A patent disclosure has been filed for the use of the Faraday forcing method on levitated liquid droplets for the measurement of interfacial tension. In the second task, the Faraday forcing method has been successfully used to measure interfacial tension in a constrained geometry. Here two liquids in a vertically stacked configuration are electrostatically forced (not levitated) until the forcing or parametric frequency excites the natural frequency. The amplitude of the forcing is measured and compared to its theoretical prediction. From this comparison the interfacial tension is determined. In a series of experiments conducted in JAXA's (Japanese Aerospace Exploration Agency) ground Electrostatic apparatus with silicone oil and water as test fluids, the theory and experiments have been compared. It was concluded from these experiments that this method gave accurate results for interfacial tension. It was further concluded that this method will be a very viable means to determine interfacial between liquid metals and encapsulants. This will be our ultimate goal. The following papers have been submitted to the noted journals: N. Brosius, K. Ward, S. Matsumoto, M. SanSoucie and R. Narayanan "Faraday forcing of high-temperature levitated liquid metal drops for the measurement of surface tension." Submitted to npg Microgravity, Mar-2018 K. Ward, S. Matsumoto and R. Narayanan "Electrostatically forced Faraday instability – theory and experiments." Submitted to Journal of Fluid Mechanics, Apr-2018
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Dissertations and Theses	Ward K. "Faraday Instability in Mechanically and Electrostatically Forced Systems." Dissertation, University of Florida, Submitted April 2018. , Apr-2018

Patents	Serial No.: 62/577,562. Pending as of October 2017. Oct-2017 Narayanan R, Ward K, Brosius N, SanSoucie M, Matsumoto S. "Interfacial tension measurement using continuously forced electrostatically levitated drops."